

Mean-Square Stability Analysis with Distributed Stochastic Uncertainty

Maurice Filo, ETH Zurich

maurice.filo@bsse.ethz.ch

Swiss Federal Institute of Technology in Zürich (ETHz)
University of California, Santa Barbara (UCSB)

<https://maurice-filo.github.io/>

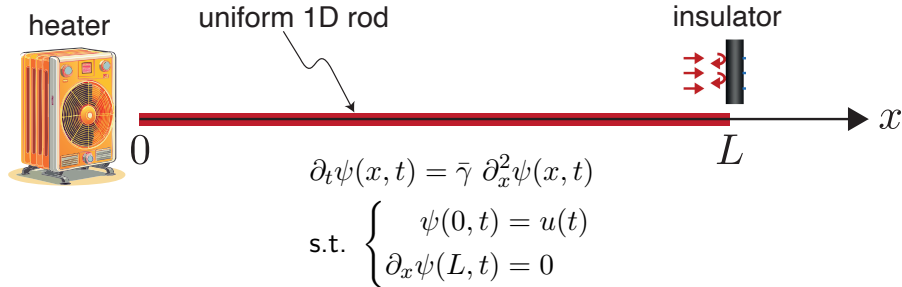


ETH zürich

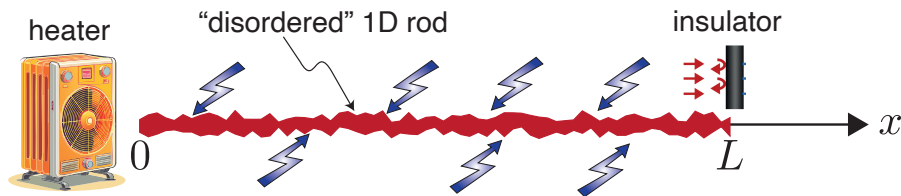
MTNS 2024, Cambridge



Example 1: Heat Equation in Disordered Material

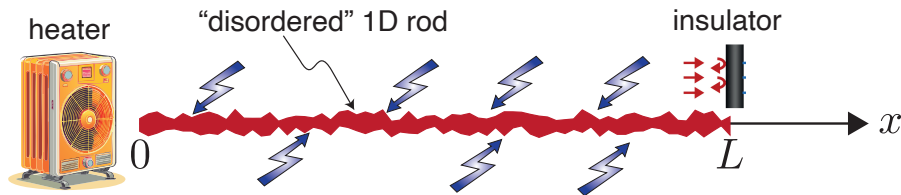


Example 1: Heat Equation in Disordered Material



$$\partial_t \psi(x, t) = \partial_x \left[\left(\bar{\gamma} + \underset{\substack{\uparrow \\ \text{multiplicative}}}{\gamma(x, t)}} \right) \partial_x \psi(x, t) \right] + \underset{\substack{\uparrow \\ \text{additive}}}{w(x, t)}$$

Example 1: Heat Equation in Disordered Material

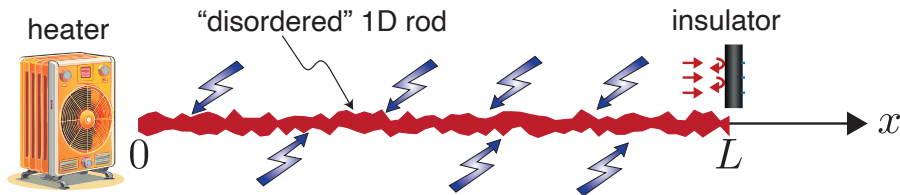


$$\partial_t \psi(x, t) = \partial_x \left[\left(\bar{\gamma} + \underset{\substack{\uparrow \\ \text{multiplicative}}}{\gamma(x, t)}} \right) \partial_x \psi(x, t) \right] + \underset{\substack{\uparrow \\ \text{additive}}}{w(x, t)}$$

$$\mathbb{E} [\gamma(x, t) \gamma(\xi, \tau)] =: \Gamma(x, \xi) \delta(t - \tau)$$

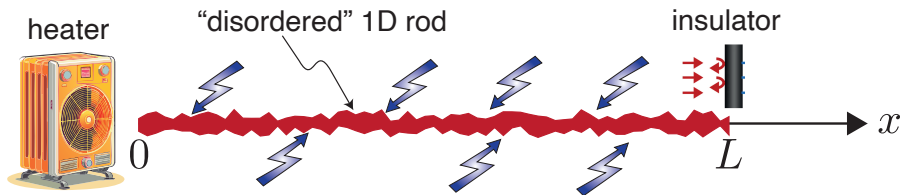
$$\mathbb{E} [w(x, t) w(\xi, \tau)] =: W(x, \xi) \delta(t - \tau)$$

Example 1: Heat Equation in Disordered Material



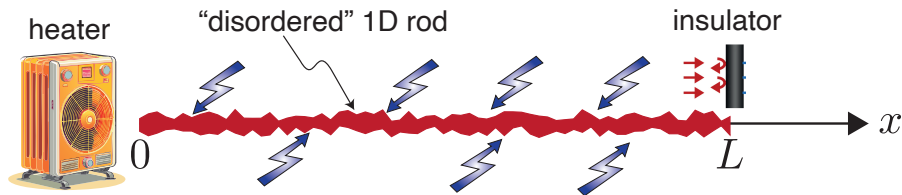
$$\begin{aligned}\partial_t \psi(x, t) &= \partial_x \left[\left(\bar{\gamma} + \gamma(x, t) \right) \partial_x \psi(x, t) \right] + w(x, t) \\ &= \bar{\gamma} \partial_x^2 \psi(x, t) + \left[\partial_x \quad \mathcal{I} \right] \gamma(x, t) \begin{bmatrix} \partial_x \\ \partial_x^2 \end{bmatrix} \psi(x, t) + w(x, t)\end{aligned}$$

Example 1: Heat Equation in Disordered Material



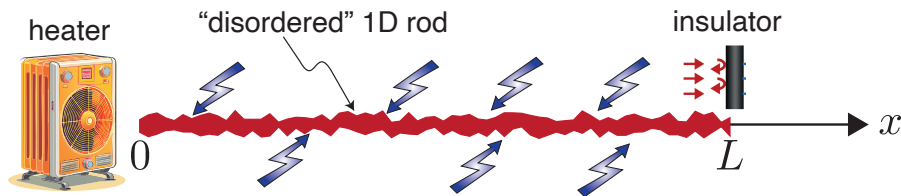
$$\begin{aligned} \partial_t \psi(x, t) &= \partial_x \left[\left(\bar{\gamma} + \gamma(x, t) \right) \partial_x \psi(x, t) \right] + w(x, t) \\ &= \underbrace{\bar{\gamma} \partial_x^2}_{A} \psi(x, t) + \underbrace{\left[\partial_x \mathcal{I} \right]}_B \gamma(x, t) \underbrace{\left[\frac{\partial_x}{\partial_x^2} \right]}_C \psi(x, t) + w(x, t) \end{aligned}$$

Example 1: Heat Equation in Disordered Material



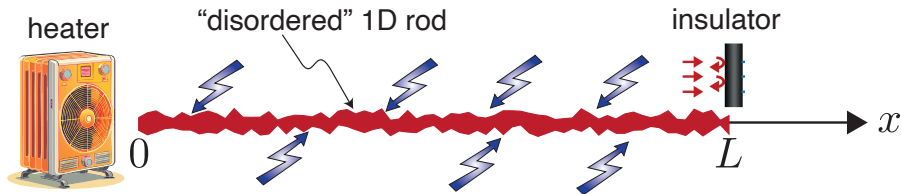
$$\partial_t \psi(x, t) = \mathcal{A} \psi(x, t) + \mathcal{B} \gamma(x, t) \mathcal{C} \psi(x, t) + w(x, t)$$

Example 1: Heat Equation in Disordered Material

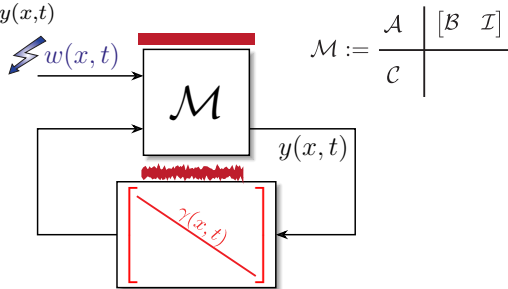


$$\partial_t \psi(x, t) = \mathcal{A} \psi(x, t) + \mathcal{B} \gamma(x, t) \underbrace{\mathcal{C} \psi(x, t)}_{y(x, t)} + w(x, t)$$

Example 1: Heat Equation in Disordered Material



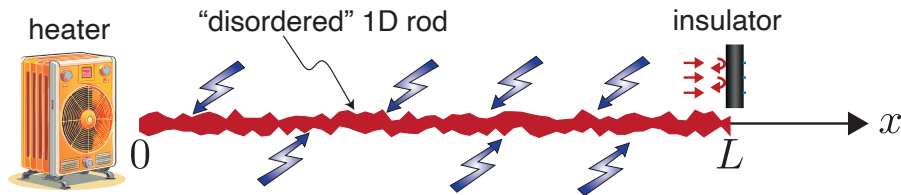
$$\partial_t \psi(x, t) = \mathcal{A} \psi(x, t) + \underbrace{\mathcal{B} \gamma(x, t) \mathcal{C} \psi(x, t)}_{y(x, t)} + w(x, t)$$



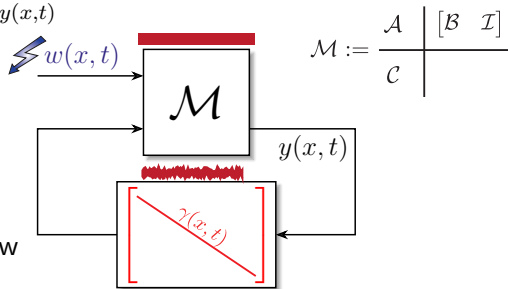
$$\mathbb{E} [\gamma(x, t) \gamma^*(\xi, \tau)] = \mathbf{\Gamma}(x, \xi) \delta(t - \tau)$$

$$\mathbb{E} [w(x, t) w^*(\xi, \tau)] = \mathbf{W}(x, \xi) \delta(t - \tau)$$

Example 1: Heat Equation in Disordered Material



$$\partial_t \psi(x, t) = \mathcal{A} \psi(x, t) + \underbrace{\mathcal{B} \gamma(x, t) \mathcal{C} \psi(x, t)}_{y(x, t)} + w(x, t)$$



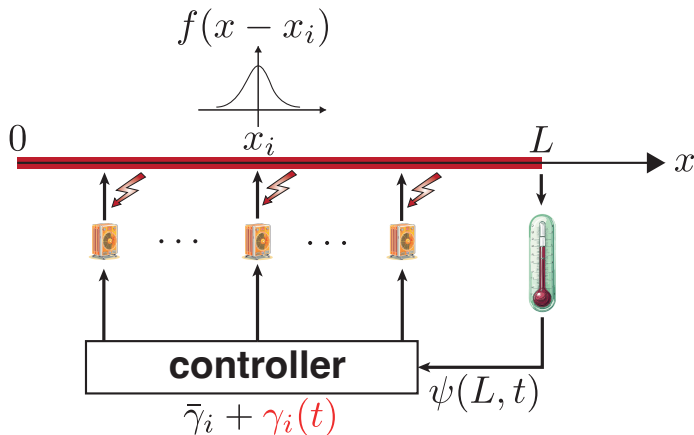
Two main questions:

- Covariance signals remain bounded? Conditions?
- If not, what spatial modes grow the fastest?

$$\mathbb{E} [\gamma(x, t) \gamma^*(\xi, \tau)] = \mathbf{\Gamma}(x, \xi) \delta(t - \tau)$$

$$\mathbb{E} [w(x, t) w^*(\xi, \tau)] = \mathbf{W}(x, \xi) \delta(t - \tau)$$

Example 2: Feedback Control



$$\partial_t \psi(x, t) = \alpha \partial_x^2 \psi(x, t) - \sum_{i=1}^n (\bar{\gamma}_i + \gamma_i(t)) f(x - x_i) \psi(L, t)$$

$$\gamma(t) := [\gamma_1(t) \quad \cdots \quad \gamma_n(t)]^T \quad \mathbb{E} [\gamma(t) \gamma^T(\tau)] =: \mathbf{\Gamma} \delta(t - \tau)$$

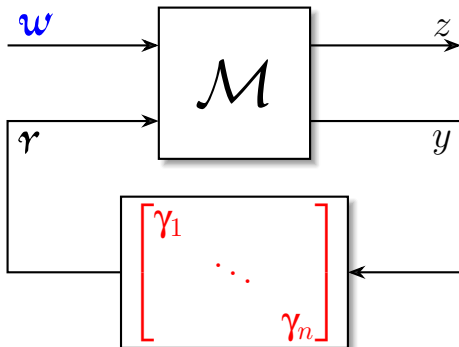
Multiple Settings

- Time of dynamics $\in \{\text{continuous, discrete}\}$
- Space of nominal dynamics $\in \{\text{continuous, discrete}\}$
- Space of structured stochastic uncertainty $\in \{\text{continuous, discrete}\}$

Overview

- Problem Statement: I/O Structured Stochastic Uncertainty
- Stochastic Block Diagrams
- Block Diagram Conversion Between Stochastic Interpretations
- Loop Gain Operator
- Conditions for Mean-Square Stability
- Application to State-Space Realizations
- Final Remarks & Conclusion

Structured Stochastic Uncertainty: Input/Output Approach

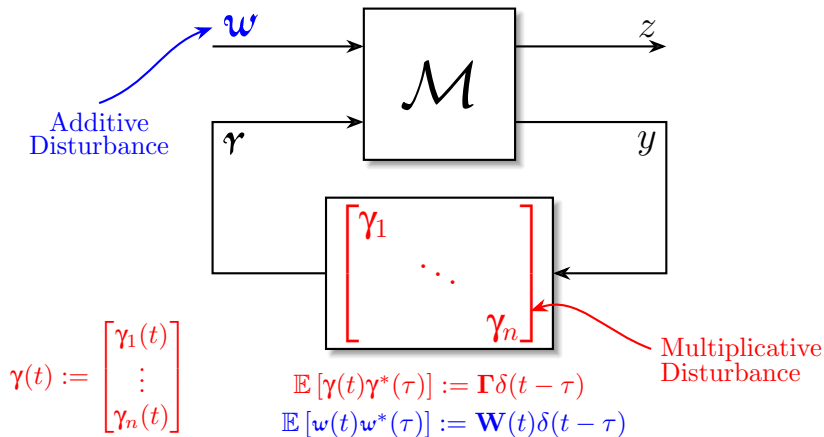


$$\gamma(t) := \begin{bmatrix} \gamma_1(t) \\ \vdots \\ \gamma_n(t) \end{bmatrix}$$

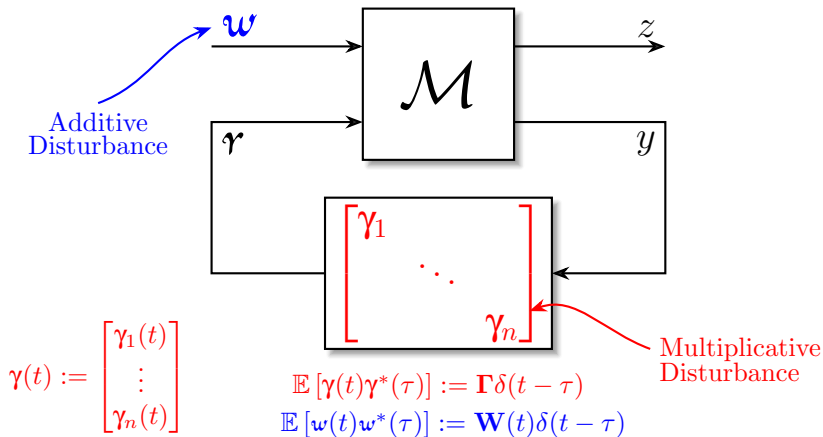
$$\mathbb{E} [\gamma(t) \gamma^*(\tau)] := \mathbf{\Gamma} \delta(t - \tau)$$

$$\mathbb{E} [w(t) w^*(\tau)] := \mathbf{W}(t) \delta(t - \tau)$$

Structured Stochastic Uncertainty: Input/Output Approach



Structured Stochastic Uncertainty: Input/Output Approach

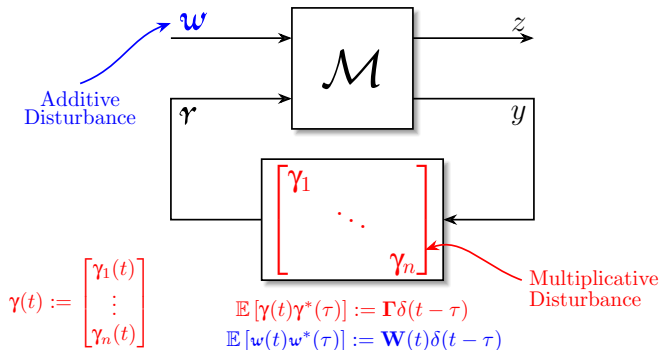


$$\begin{bmatrix} z \\ y \end{bmatrix} = \mathcal{M} \begin{bmatrix} w \\ r \end{bmatrix} \iff \begin{bmatrix} z(t) \\ y(t) \end{bmatrix} = \int_0^t M(t - \tau) \begin{bmatrix} w(\tau) \\ r(\tau) \end{bmatrix} d\tau$$

$$r(t) = \text{Diag}(\gamma(t))y(t).$$

Mean-Square Stability & Structured Stochastic Uncertainty

Goal: What are the conditions of MSS?

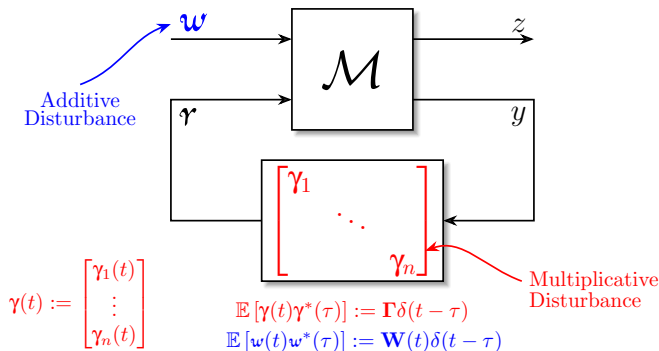


$$\begin{bmatrix} z \\ y \end{bmatrix} = \mathcal{M} \begin{bmatrix} \mathbf{w} \\ \mathbf{r} \end{bmatrix} \iff \begin{bmatrix} z(t) \\ y(t) \end{bmatrix} = \int_0^t M(t - \tau) \begin{bmatrix} \mathbf{w}(\tau) \\ \mathbf{r}(\tau) \end{bmatrix} d\tau$$

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Mean-Square Stability & Structured Stochastic Uncertainty

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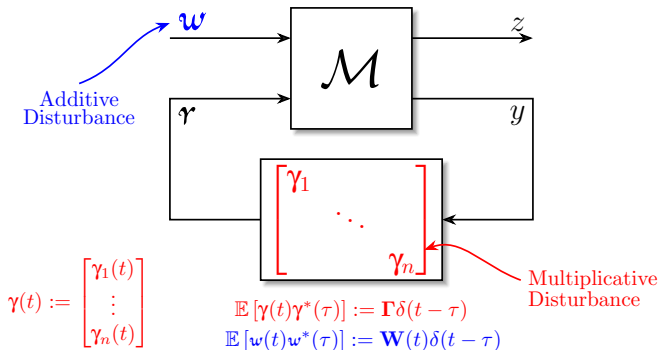


$$\begin{bmatrix} z \\ y \end{bmatrix} = \mathcal{M} \begin{bmatrix} \mathbf{w} \\ \mathbf{r} \end{bmatrix} \iff \begin{bmatrix} z(t) \\ y(t) \end{bmatrix} = \sum_{\tau=0}^t M(t-\tau) \begin{bmatrix} \mathbf{w}(\tau) \\ \mathbf{r}(\tau) \end{bmatrix}$$

$$\mathbf{r}(t) = \text{Diag}(\boldsymbol{\gamma}(t))\mathbf{y}(t).$$

Mean-Square Stability & Structured Stochastic Uncertainty

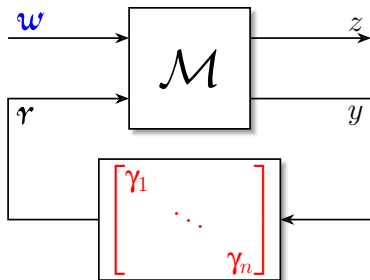
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$$\begin{bmatrix} z \\ y \end{bmatrix} = \mathcal{M} \begin{bmatrix} \mathbf{w} \\ \mathbf{r} \end{bmatrix} \iff \begin{bmatrix} z(t) \\ y(t) \end{bmatrix} = \int_0^t M(t-\tau) \begin{bmatrix} \mathbf{w}(\tau) \\ \mathbf{r}(\tau) \end{bmatrix} d\tau$$

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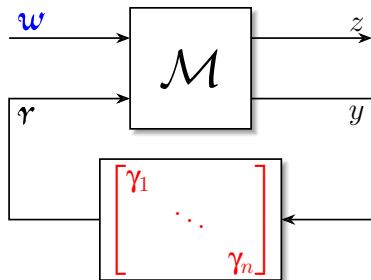
Stochastic Block Diagrams



$$\mathbb{E} [\gamma(t)\gamma^*(\tau)] = \mathbf{\Gamma}\delta(t - \tau)$$

$$\mathbb{E} [w(t)w^*(\tau)] = \mathbf{W}(t)\delta(t - \tau)$$

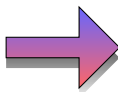
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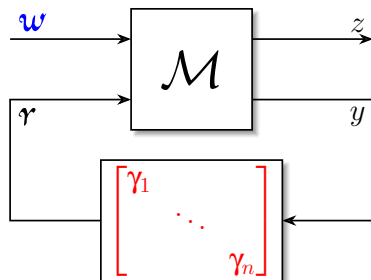
$$w := \frac{dw}{dt}$$



$$\gamma := \frac{d\gamma}{dt}$$

$$r := \frac{dr}{dt}$$

Stochastic Block Diagrams



$$\mathbb{E}[\gamma(t)\gamma^*(\tau)] = \mathbf{\Gamma}\delta(t - \tau)$$

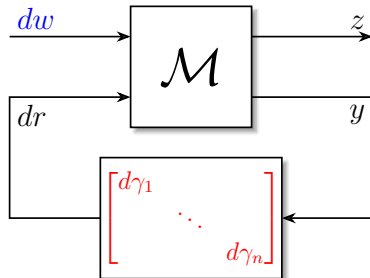
$$\mathbb{E}[w(t)w^*(\tau)] = \mathbf{W}(t)\delta(t - \tau)$$

White Process Representation

$$w := \frac{dw}{dt}$$

$$\gamma := \frac{d\gamma}{dt}$$

$$r := \frac{dr}{dt}$$



$$\mathbb{E}[d\gamma(t)d\gamma^*(t)] = \mathbf{\Gamma}dt$$

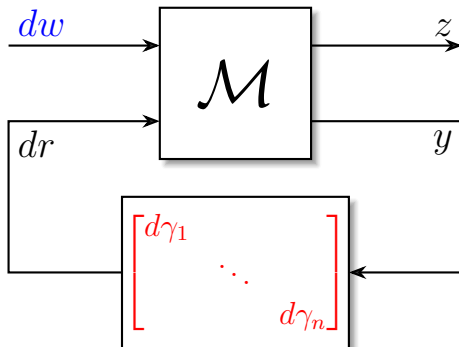
$$\mathbb{E}[dw(t)dw^*(t)] = \mathbf{W}(t)dt$$

Wiener Process Representation

$$\begin{bmatrix} z \\ y \end{bmatrix} = \mathcal{M} \begin{bmatrix} dw \\ dr \end{bmatrix} \iff \begin{bmatrix} z(t) \\ y(t) \end{bmatrix} = \int_0^t M(t - \tau) \begin{bmatrix} dw(\tau) \\ dr(\tau) \end{bmatrix}$$

$$dr(t) = \text{Diag}(d\gamma(t))y(t).$$

Stochastic Interpretations: Itô & Stratonovich



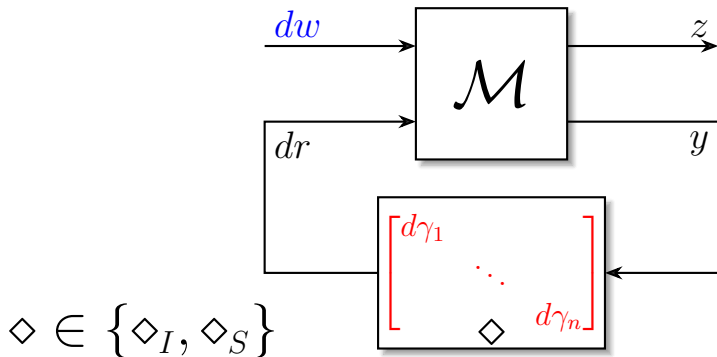
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$$dr(t) = \text{Diag}(\textcolor{red}{d}\boldsymbol{\gamma}(t))y(t).$$

Stochastic Interpretations: Itô & Stratonovich



$$\diamond \in \{\diamond_I, \diamond_S\}$$

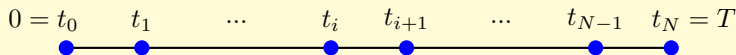
$$\mathbb{E}[d\gamma(t)d\gamma^*(t)] = \mathbf{\Gamma}dt$$

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$$dr(t) = \text{Diag}(\textcolor{red}{d\gamma}(t)) \diamond y(t).$$

Recall: Stochastic Integrals



- **Deterministic:**

$$\int_0^T v(t) dt := \lim_{N \rightarrow \infty} \sum_{k=0}^{N-1} v(\bar{t}_i) (t_{i+1} - t_i); \quad \forall \bar{t}_i \in [t_i, t_{i+1}]$$

- **Itô:**

$$\int_0^T v(t) \diamond_I dw(t) := \lim_{N \rightarrow \infty} \sum_{i=0}^{N-1} v(t_i) (w(t_{i+1}) - w(t_i))$$

- **Stratonovich:**

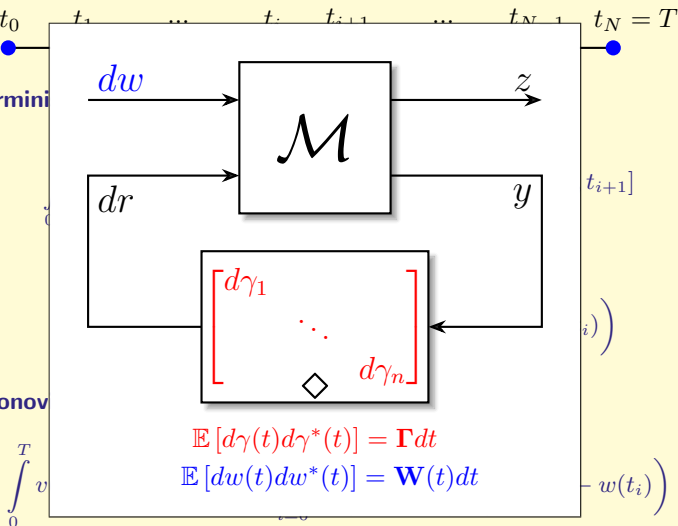
$$\int_0^T v(t) \diamond_S dw(t) := \lim_{N \rightarrow \infty} \sum_{i=0}^{N-1} v\left(\frac{t_i + t_{i+1}}{2}\right) (w(t_{i+1}) - w(t_i))$$

Recall: Stochastic Integrals

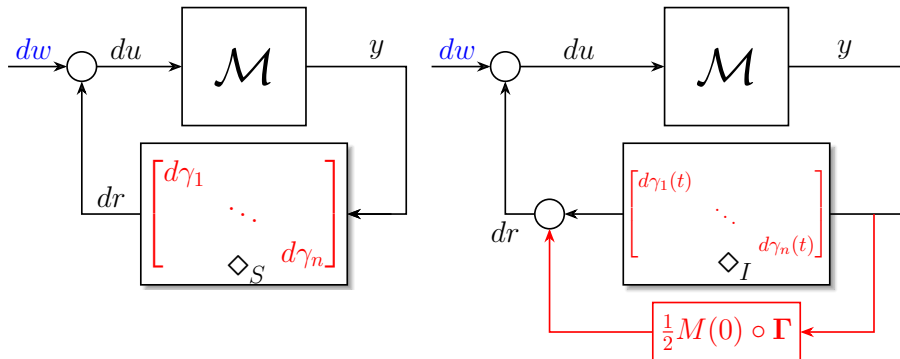
• Deterministic

• Itô:

• Stratonovich



Stratonovich to Ito Conversion



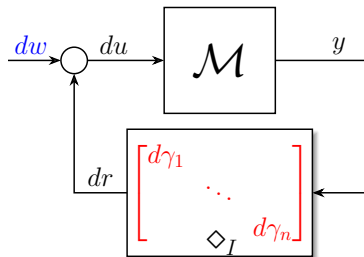
$$\mathbb{E} [d\gamma(t)d\gamma^*(t)] := \mathbf{\Gamma} dt;$$

$$y(t) = \int_0^t M(t - \tau) du(\tau);$$

“ $\circ$ ” is the Hadamard (element-by-element) product

The two stochastic block diagrams are “equivalent in the mean-square sense”.

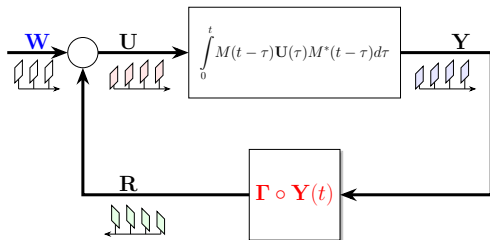
Loop Gain Operator & Mean-Square Stability



$$\mathbb{E}[d\gamma(t)d\gamma^*(t)] = \mathbf{\Gamma}dt$$

$$\mathbb{E}[dw(t)dw^*(t)] = \mathbf{W}(t)dt$$

Stochastic Block Diagram

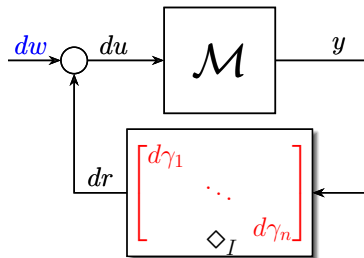


$$\mathbb{E}[du(t)du^*(t)] = \mathbf{U}(t)dt; \quad \mathbb{E}[y(t)y^*(t)] = \mathbf{Y}(t);$$

$$\mathbb{E}[dr(t)dr^*(t)] = \mathbf{R}(t)dt; \quad \text{"} \circ \text{"}: \text{Hadamard Product};$$

Deterministic Covariance Block Diagram

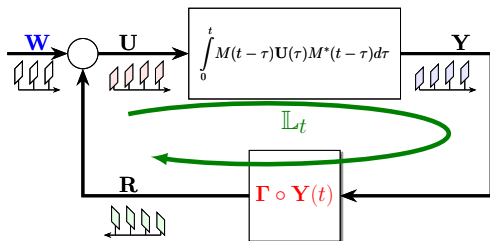
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Stochastic Block Diagram



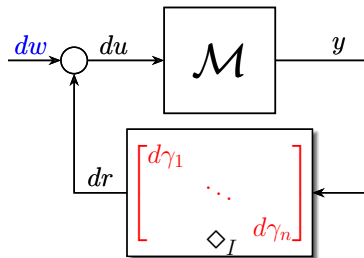
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Deterministic Covariance Block Diagram

$$\mathbb{L}_t(\mathbf{U}) := \mathbf{\Gamma} \circ \left(\int_0^t M(t-\tau)\mathbf{U}(\tau)M^*(t-\tau)d\tau \right), \quad \mathbb{L} := \lim_{t \rightarrow \infty} \mathbb{L}_t$$

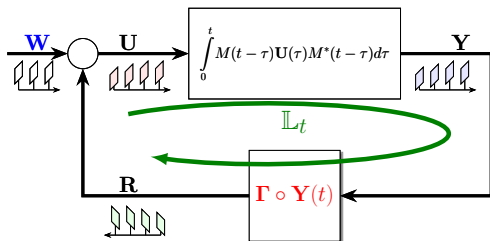
Loop Gain Operator & Mean-Square Stability



$$\mathbb{E}[d\gamma(t)d\gamma^*(t)] = \mathbf{\Gamma}dt$$

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Stochastic Block Diagram



$$\mathbb{E}[du(t)du^*(t)] = \mathbf{U}(t)dt; \quad \mathbb{E}[y(t)y^*(t)] = \mathbf{Y}(t);$$

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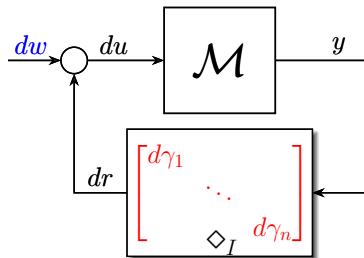
Deterministic Covariance Block Diagram

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Necessary & Sufficient Conditions of Mean-Square Stability:

- Forward Block is Stable (Finite H^2 -norm)
- Spectral Radius of $\mathbb{L}$ is strictly less than 1, $\rho(\mathbb{L}) < 1$

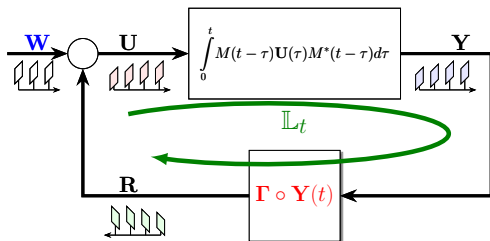
Loop Gain Operator & Mean-Square Stability



$$\mathbb{E}[d\gamma(t)d\gamma^*(t)] = \mathbf{\Gamma}dt$$

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Stochastic Block Diagram



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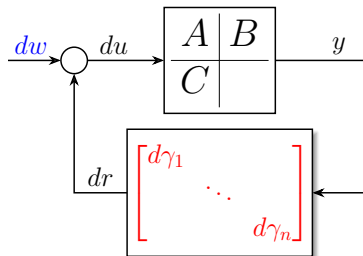
Deterministic Covariance Block Diagram

$$\mathbb{L}_t(\mathbf{U}) := \mathbf{\Gamma} \circ \left(\int_0^t \mathbf{M}(t-\tau)\mathbf{U}(\tau)\mathbf{M}^*(t-\tau)d\tau \right), \quad \mathbb{L} := \lim_{t \rightarrow \infty} \mathbb{L}_t$$

Two important quantities related to $\mathbb{L}$:

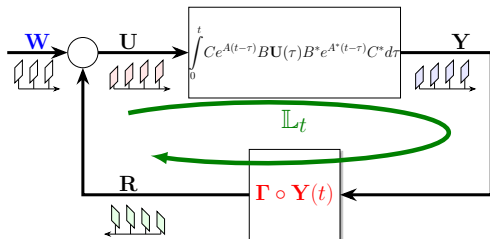
- Spectral Radius: $\rho(\mathbb{L})$
- Worst-Case Covariance: $\mathbb{L}(\hat{\mathbf{U}}) = \rho(\mathbb{L})\hat{\mathbf{U}}$ (Perron-Frobenius "Eigen-matrix")

Application to State-Space Realizations



$$\begin{aligned}\mathbb{E}[d\gamma(t)d\gamma^*(t)] &= \mathbf{\Gamma}dt \\ \mathbb{E}[dw(t)dw^*(t)] &= \mathbf{W}(t)dt\end{aligned}$$

Stochastic Block Diagram

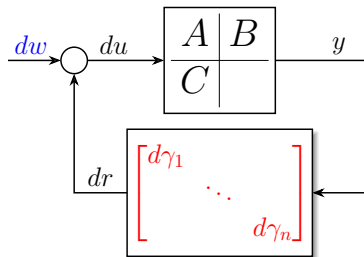


$$\begin{aligned}\mathbb{E}[du(t)du^*(t)] &= \mathbf{U}(t)dt; & \mathbb{E}[y(t)y^*(t)] &= \mathbf{Y}(t); \\ \mathbb{E}[dr(t)dr^*(t)] &= \mathbf{R}(t)dt; & \text{"}\circ\text{"} &: \text{Hadamard Product};\end{aligned}$$

Deterministic Covariance Block Diagram

$$\mathbb{L}_t(\mathbf{U}) := \mathbf{\Gamma} \circ \left(\int_0^t C e^{A(t-\tau)} B \mathbf{U}(\tau) B^* e^{A^*(t-\tau)} C^* d\tau \right), \quad \mathbb{L} := \lim_{t \rightarrow \infty} \mathbb{L}_t$$

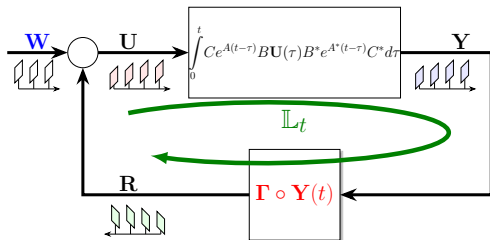
Application to State-Space Realizations



$$\mathbb{E}[d\gamma(t)d\gamma^*(t)] = \mathbf{\Gamma}dt$$

$$\mathbb{E}[dw(t)dw^*(t)] = \mathbf{W}(t)dt$$

Stochastic Block Diagram



$$\mathbb{E}[du(t)du^*(t)] = \mathbf{U}(t)dt; \quad \mathbb{E}[y(t)y^*(t)] = \mathbf{Y}(t);$$

$$\mathbb{E}[dr(t)dr^*(t)] = \mathbf{R}(t)dt; \quad \text{"} \circ \text{"}: \text{Hadamard Product};$$

Deterministic Covariance Block Diagram

$$\mathbb{L}_t(\mathbf{U}) := \mathbf{\Gamma} \circ \left(\int_0^t C e^{A(t-\tau)} B \mathbf{U}(\tau) B^* e^{A^*(t-\tau)} C^* d\tau \right), \quad \mathbb{L} := \lim_{t \rightarrow \infty} \mathbb{L}_t$$

$$\bar{\mathbf{R}} = \mathbb{L}(\bar{\mathbf{U}}) \quad \Leftrightarrow \quad \begin{cases} \bar{\mathbf{R}} = \mathbf{\Gamma} \circ (C \bar{\mathbf{X}} C^*) \\ 0 = A_k \bar{\mathbf{X}} + \bar{\mathbf{X}} A_k^* + B \bar{\mathbf{U}} B^* \end{cases}$$

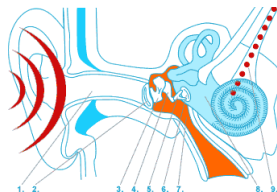
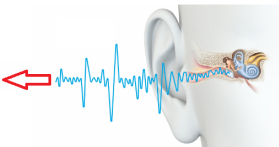
$$\text{where } A_{\text{It}\bar{o}} := A \text{ and } A_{\text{Strat}} := A + \frac{1}{2} B ((CB) \circ \mathbf{\Gamma}) C$$

Proofs & An Application

- **Discrete-Time Setting:** Bamieh, B., & Filo, M. (2020). An input–output approach to structured stochastic uncertainty. *IEEE Transactions on Automatic Control*.
- **Continuous-Time Setting:** Filo, M., & Bamieh, B. (2018). An input-output approach to structured stochastic uncertainty in continuous time. *arXiv preprint*.
- **Stochastic Block Diagrams:** Filo, M., & Bamieh, B. (2018). A block diagram approach to stochastic calculus with application to multiplicative uncertainty analysis. *IEEE CDC*.

Proofs & An Application

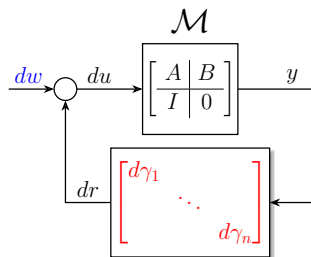
- **Discrete-Time Setting:** Bamieh, B., & Filo, M. (2020). An input–output approach to structured stochastic uncertainty. IEEE Transactions on Automatic Control.
- **Continuous-Time Setting:** Filo, M., & Bamieh, B. (2018). An input-output approach to structured stochastic uncertainty in continuous time. arXiv preprint.
- **Stochastic Block Diagrams:** Filo, M., & Bamieh, B. (2018). A block diagram approach to stochastic calculus with application to multiplicative uncertainty analysis. IEEE CDC.
- **Application:** Stochastic Instabilities in the inner ear!



→ Filo, M., & Bamieh, B. (2017). Investigating cochlear instabilities using structured stochastic uncertainty. IEEE CDC

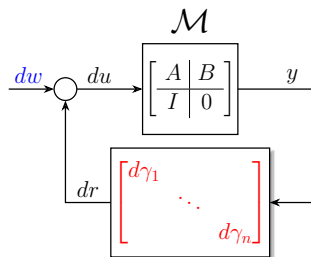
→ Filo, M., & Bamieh, B. (2020). Investigating Instabilities in the Mammalian Cochlea Using a Stochastic Uncertainty Model. IEEE Transactions on Molecular, Biological, and Multi-Scale Communications.

Concluding Remarks & Future Work



$$\text{SDE: } dy(t) = Ay(t)dt + B\text{Diag}\left(d\gamma(t)\right)y(t) + Bdw(t)$$

Concluding Remarks & Future Work

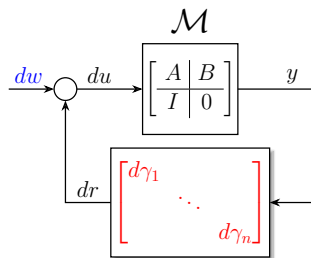


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Extends and unifies the analysis for systems $\mathcal{M}$:

- State space realizations
- Infinite dimensional systems with finite number of multiplicative disturbances
- Systems with delays

Concluding Remarks & Future Work



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Extends and unifies the analysis for systems $\mathcal{M}$:

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Future Direction: Extend the analysis for

- *Colored* disturbances
- Spatially distributed disturbances with symmetries.